MIL-B-17089A(Aer)
18 April 1957
Supersedes MIL-B-17089(Aer)
of 14 March 1952

MILITARY SPECIFICATION
BAROMETER, ANEROID ML-448( )/UM

This specification has been approved
by the Bureau of Aeronautics, Department of the Navy

1. SCOPE AND CLASSIFICATION

1.1 Scope.--This specification covers aneroid barometers which are for use by Navy Weather activities to measure atmospheric air pressure by movements of an evacuated metal diaphragm or diaphragms for pressures in the range of 910 to 1060 millibars.

1.2 Type.--This specification covers one type of Barometer, designated ML-448( )/UM.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, publications and standards of the issue in effect on date of invitation for bids form a part of this specification.

SPECIFICATIONS:

Federal

QQ-B-621 - Brass, Commercial Yellow, High-Copper-Yellow and Naval: castings

QQ-B-691 - Bronze, castings

Military

MIL-R-900 - Rubber, Synthetic, Medium-soft, Molded Sheet, and Strip (for Airport Hatch and Watertight and Airtight Door Gaskets)

MIL-E-2036 - Enclosures (For Non-rotating Electrical Equipment) (Naval Shipboard Use)

MIL-P-6064 - Packaging of Light-weight Aircraft Accessories

MIL-S-6872 - Soldering Process, General Specifications for

MIL-E-15090 - Enamel, Equipment, Light-gray

MIL-N-18307(Aer) - Nomenclature and Nameplates for Airborne Electronic and Associated Equipment

STANDARDS:

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-125 - Manual of Standard Descriptions for Electronic Equipment and Material

MIL-STD-129 - Marking of Shipments

(When requesting specifications, publications and drawings refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania, Attention: Code ODPT.)

2.2 Other Publications.

National Bureau of Standards

Handbook H28 - Screw thread Standards for Federal Service.

(Copies of Handbook H28 may be obtained upon application, accompanied by money order, coupon or cash, to Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

FSC 6685

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3. REQUIREMENTS

3.1 Component Parts.--The barometer shall consist essentially of a case, crystal, bezel ring, spacer ring, dial with mirror ring, index hand, and aneroid mechanism.

3.2 Preproduction Samples.--Preproduction samples will be required by this specification. They shall be submitted to the Government agency concerned.

3.2.1 Samples.--After award of contract, the contractor shall proceed with fabrication of and shall assemble samples of barometer, using tools and methods which are the same, or representative of those to be used in quantity production. Two (2) samples shall be submitted for all tests of this specification and other tests not specifically described herein when such tests are deemed necessary to determine full compliance with the specifications.

3.2.2 Sample Shipment.--The samples shall be marked and shipped in accordance with directions given in the contract and specification.

3.2.3 Approval.--Approval of the submitted samples shall not be construed as authorizing any deviations from the requirement of this specification. If any sample is disapproved, the contractor may be required to submit additional samples until the required quantity of satisfactory samples has been submitted. Such additional samples shall be accompanied by a description of the changes made to correct the faults of the rejected samples.

3.3 Service Condition Requirement.--The equipment shall be capable of satisfactory operation after exposure to, or storage under, any possible combination of the service conditions listed below, and shall be capable of satisfactory operation under those conditions where specifically indicated.

a. Temperature - Exposure and/or Storage: -50°F to +160°F
   - Operation -20°F to +120°F

b. Relative Humidity - Operation, Exposure and/or Storage: Any relative humidity up to 100%, including condensation caused by temperature changes.

c. Fungi and Insects - As encountered in tropical service.

d. Pressure - Withstand pressures down to 500 millibars and up to 1150 millibars.

e. Vibration - Vibration as encountered aboard ship and in aircraft transportation as specified in Paragraph 4.2.16.

f. Gun Blast Proof - Gun Blast Proof as required in Paragraph 4.2.17.

g. Salt-laden Air - Operation, Exposure and/or Storage: As encountered aboard ship.

In addition, the equipment shall be capable of meeting the tests in Section 4.

3.4 Material.

3.4.1 Material.--The best material for Tropical and Arctic use commercially available for the purpose shall be used. Wherever possible, metals used shall be non-corroding, if corrodible metals are used they shall, wherever possible, be adequately protected from corrosion by plating, lacquer, baked enamel, or other suitable covering or finish; finishes which will crack, chip or peel under normal usage and aging or when subjected to changes of temperature and humidity shall not be employed. Wear surfaces of gear teeth, pivots, bearing faces, etc., which cannot be protected with finishes, shall be made of a material which will not appreciably tarnish or form oxidation films.

3.4.2 Case, Bezel Ring, and Spacer Ring.--The case shall be of cast, spun or fabricated construction. Aluminum is not acceptable. Brass conforming to Federal Specification QQ-B-621 or Bronze conforming to Federal Specification QQ-B-691 are suitable materials. A bezel ring of the same material as the case, and a spacer ring of white enameled brass shall be provided.

3.4.3 Dial.--The dial shall be of metal. The minimum acceptable over-all diameter shall be 5 inches.

3.4.4 Crystal.--The crystal shall be of heat-treated glass, capable of withstanding a pressure differential between the outside and the inside of the case of 500 millibars, and gun blast as specified in Paragraph 4.2.17. It shall be flat, clear, and free of bubbles or flaws.

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3.4.5 Undesignated parts.--Undesignated parts shall be of non-corrosive materials. No two metals shall be used in contact if their characteristics are such that contact between them results in accelerated corrosion in one of the materials.

3.4.6 Screws.--All screws, nuts, and other similar parts used in the assembly of the instrument shall conform to the National Bureau of Standards Handbook H28, and shall be Class 2 fit unless otherwise specified herein. In general screws in hard material shall be fine thread series. All screws, nuts, bolts and washers shall be protected against corrosion.

3.4.6.1 Set Screws.--Set screws shall be of the hardened cup-pointed type.

3.4.7 Critical Material.--The contractor may substitute non-critical materials for critical materials during national emergencies upon receiving approval of the contracting agency.

3.5 Detailed Construction Requirements.--The barometer shall be so constructed that no parts are likely to work loose in service. The strains, impact, shocks, vibrations, and such other conditions incident to usual shipping, storage, installation, and service at sea shall not affect its serviceability nor accuracy.

3.5.1 Case.--The case shall be well and substantially made. It shall be circular in shape and shall have an inside depth of not less than 2 inches nor more than 4 inches. It shall have a diameter across the body of not less than 6 inches nor more than 8 inches. A circular mounting flange shall be an integral part of the case. The flange shall be sturdily made and the diameter shall not exceed that of the case by more than 1 inch. Three mounting holes shall be spaced at uniform distances around the flange. They shall be so located that the barometer can be mounted on a wall or similar support without the use of special tools. The top hole shall be approximately in line with the 1015 mb. mark on the dial. The case shall have a rubber mounting pad attached to the rear of the mounting flange of sufficient size to provide a cushion mounting. A removable mounting base of a design suitable to permit standing the barometer on a horizontal surface, shall be attached to the case. The zero adjusting screw shall be so located as to be accessible through the bottom of the case. The interior of the case shall provide for the mounting of the frame to which the movement is attached. The design shall be such that the frame and attached movement may be easily removed for purposes of calibration and adjustment. The exterior of the case shall be grain polished and protected by a transparent lacquer or plastic. A vent hole for passage of air shall be provided in the lower wall of the case. This hole shall be threaded for 1/8" pipe, and shall be closed with a removable brass plug having a small vent hole. When assembled for normal use, and with an air-tight seal provided over the vent hole, the instrument shall be capable of withstanding a pressure differential of 500 millibars and the leakage rate shall be satisfactory to the Bureau of Aeronautics (AT-2).

3.5.1.1 Bezel Ring.--The bezel ring shall have the same finish as the case. The ring shall be approximately 7/8 inch wide and sufficiently thick to be rigid. The bezel ring shall be attached to the case by means of threads and it shall be so constructed such that after its removal, the removal of the barometer movement from the case is permitted. The bezel ring shall fit against the crystal sufficiently tight to limit leakage rate as in 3.5.1 above. The rubber gasket used thereupon shall be in accordance with Specification MIL-R-900.

3.5.1.2 Spacer Ring.--The Spacer Ring shall be 0.015 to 0.030 inch thick. The outside diameter shall be slightly smaller than the inside diameter of the case.

3.5.2 Dial.--The dial shall have a dull opaque, white finish. All lettering, numerals, and graduations shall be printed, engraved or stamped on the surface and shall be colored black. The dial shall be provided with two concentric scales, a scale graduated in inches of mercury and scale graduated in millibars. The outer scale shall have a range of from 910 to 1060 millibars and the inner scale from 26.87 to 31.3 inches of mercury. The legends "Inches of Mercury" and "Millibars" in 1/16 inch letters shall be marked on the dial so located as to identify the scale to which they apply. The scales shall be uniform and substantially linear. The indications of the inch scale shall be equivalent to the corresponding values in the millibar scale over the entire range of scale. The inches of mercury scale shall be graduated in inches, tenths and two-hundredths, and the millibar scale shall be graduated in millibars and 1/2 millibars. Concentric with and between the scales shall be a mirror ring not less than 1/8 inch wide. The mirror surface shall be permanent without tendency to tarnish or discolor.

3.5.2.1 The length of scale graduations shall be approximately as specified in Table I.

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Table I - Length of scale graduations

Graduation	Length of Line
	Inch
1.0 inch	5/32
0.1 inch	1/8
0.02 inch	1/16
10 millibar	9/64
5 millibar	7/64
1.0 millibar	5/64
0.5 millibar	3/64

3.5.2.2 Graduations shall be numbered with figures according to Table II

Table II - Graduations and size of figures

Graduation	Size of Figure
	Inch
1.0 inch	.125
0.1 inch	.062
10 millibar (1000, 1010, etc.)	.093
5 millibar (995, 1005, etc.)	.062

3.5.2.3 The width of all the graduations of both the inch and the millibar scale shall be .007 ±.0015 inches. The graduations of the millibar scale shall extend over not less than 335° nor more than 360° of arc. The diameter of the inner scale measured to the base line of the graduations shall be not less than 4 inches. The 1015mb mark shall be in the center uppermost position. A means shall be provided to prevent the dial from shifting its position with respect to the mechanism. The dial shall be easily removed without the use of special tools. The dial shall be made so as not to warp, peel, or change color during ordinary use such as to interfere with the operation, calibration, or reading of the instrument. The dial lettering shall be as specified in 3.5.2.2. A drawing of the proposed dial shall be submitted to procuring agency for design approval prior to the manufacture of any production quantity of dials.

3.5.3 Index Hand.--The Index Hand shall be of friction type and of rigid metal with a black lacquer finish. It shall be centered with relation to the dial and dial graduations. The hand shall be not more than the width of a graduation at the end extending over the graduations on the two scales. The end of the pointer shall extend more than half way over the smallest graduations of the outer scale but not beyond the graduation of the outer scale.

3.5.4 Weight.--Light weight is a desirable feature in the construction of the instrument described but it shall not be attained at any sacrifice of the accuracy or permanence of calibration.

3.5.5 Aneroid Mechanism.--The aneroid mechanism shall consist of an evacuated cell sensitive to variations in atmospheric pressure. It shall be connected to an indicating hand by a lever system in such a manner that the movement of the cell, for a given change in atmospheric pressure, will be magnified by the lever system and transmitted to the indicating hand with a minimum of friction in the moving parts and a minimum of distortion other than that intentionally introduced to obtain desired calibration characteristics. Mechanical stops shall be provided to prevent damage when the mechanism is exposed to pressures as specified in Paragraph 3.3.

3.5.5.1 Temperature Compensation.--The instrument shall be compensated for temperature changes as required in Paragraph 4.2.14 by the selection of materials and intrinsic design.

3.5.5.2 Adjustment Mechanism.--Means shall be provided for adjusting the pressure-sensitive cell in such manner as to enable the indicating hand to be set to read current atmospheric pressures. The structure of these adjusting means shall be such that the adjustment will not affect the calibration of the instrument. The construction of the adjusting mechanism shall be such that very small changes in the position of the indicating hand can be made accurately. The structure of this mechanism shall be such that it can be operated from outside the case with a screw driver. The access shall permit making adjustments with a screw driver with a 5/32 inch blade. Means which require the position of the instrument to be changed from normal vertical position during adjustment will not be acceptable. The width of the slot in the adjusting screw shall be at least 0.030 inches. Construction shall be such that the adjustment cannot be changed unintentionally or by vibration of Paragraph 4.2.16.

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3.5.5.3 Lever System.--The lever mechanism shall transmit the motion of the cell, suitably magnified, to the indicating hand. The structure of the lever system shall be such that the response of the cell will be transmitted to the hand with a minimum of distortion.

3.5.5.3.1 Lever System Construction.--The lever system shall be such as to produce a movement with a minimum of friction, looseness of joints, and backlash. The use of flexible chain will not be acceptable.

3.5.5.4 Position Error.--The design of the lever system shall be balanced so that the errors in indications of the instrument due to the position of the instrument shall not be greater than the position compensation tolerances given in Paragraph 4.2.8. The instrument shall be calibrated and tested with the indicating dial in a vertical position and with 1015 millibar graduation positioned approximately at the top of the instrument except during the position error test in Paragraph 4.2.8.

3.6 Drift.--The contractor shall guarantee that the calibration at the end of the first six months shall agree with the original calibration within 1.4mb at all test points over the range of the scales. The change in calibration for the following six month period shall not exceed 0.7mbs. Acceptance will be based on the manufacturer's final calibration, but instruments found by the Government not to meet the requirements herein stated at the ends of the 6 and 12 month periods, respectively, will be returned to the manufacturer for repair or replacement without additional cost to the Government including transportation charges.

3.7 Nomenclature and Nameplates.--Nomenclature shall be obtained in accordance with the provisions of Specification MIL-N-18307(Aer) and Nameplates where used shall conform thereto, including as much of the data required therein for a major unit of non-airborne equipment (see Fig. 3 thereof) as is applicable to the barometer. A government stock number to be assigned later shall also be included on the same line with the serial number. In using Specification MIL-N-18307(Aer) the Aerology Division (AY-2) of the Bureau of Aeronautics shall be substituted in lieu of the Electronics Division (EL-419) wherever it appears.

3.7.1 Nameplate Data.--Nameplate data as required above shall be marked on the face of the dial in an arrangement appropriate to the space available. A drawing of dial markings shall be submitted to the Chief, Bureau of Aeronautics (AY-2) for approval prior to manufacture of the dials.

3.7.2 Nameplate data shall also be molded or imbedded on the back of the instrument case, or placed on a nameplate as provided above, attached to the back of the case.

3.8 Workmanship.--All parts shall be manufactured and finished in a thoroughly workmanlike manner and in accordance with the best commercial practice for the manufacture of precision instruments. All dimensions shall be held as close as is consistent with good manufacturing practice. All parts shall fit and operate in a first class workmanlike manner.

3.8.1 Drawings.--After award of contract and before manufacturing the preproduction barometers, the contractor shall furnish the contracting officer a drawing or drawings showing the general design of the instrument and listing the component parts which are to be used in manufacture. The list shall indicate the materials to be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests.--The inspection and testing of barometers shall be classified as follows:

4.1.1 Preproduction Tests.--Preproduction tests are those tests listed below which will be performed on the samples submitted to determine if the samples comply with the requirements of this specification.

4.1.2 Acceptance Tests.--All routine tests performed on production units to determine compliance with this specification for acceptance under contract.

4.1.2.1 Test Order and Classification Table.--Tests are classified in accordance with the foregoing definitions as shown in Table III below and shall be performed in the order listed.

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Table III

Inspection or Test	Paragraph Reference	Preproduction	Production Acceptance Inspection
Visual and Mechanical	4.2.5	X	X
Scale Calibration at Room Temperature	4.2.6	X	X
Movement Test	4.2.7	X	X
Position Error Test	4.2.8	X	X
Temperature Cycle Test	4.2.9	X	
Pressure Cycle Test	4.2.10	X	
Friction Test	4.2.11	X	X
Hysteresis	4.2.12	X	X
Recalibration Test	4.2.13	X	X
Temperature Compensation	4.2.14	X	X
Salt Spray Corrosion Tests	4.2.15	X	
Vibration Test	4.2.16	X	
Gun Blast Proof Test	4.2.17	X	
Performance Recheck	4.2.18	X	

4.1.3 Inspection Procedures.--Samples shall be selected in accordance with the following tables and applicable provisions of MIL-STD-105. They shall be subjected to applicable tests listed in Table III, and accepted or rejected in accordance with the following procedures.

4.1.3.1 Lot Definition.--For purposes of sampling and test, a lot shall consist of all barometers offered for Government inspection at one time.

4.1.3.2 Preproduction Samples.--Two (2) preproduction samples shall be provided for preproduction tests.

4.1.3.3 Visual and Mechanical.--Sampling for Visual and Mechanical Inspection of Paragraph 4.2.5 shall be in accordance with Table IV below.

Table IV

Lot Size (number of Barometers)	Sample Size (number of Barometers)	Acceptance number (defective)	Rejection number (defective)
9 to 12	8	0	1
13 to 17	10	0	1
18 to 25	13	0	1
26 to 35	17	1	2
36 to 50	22	1	2
51 to 70	28	1	2
71 to 100	35	2	3

4.1.3.4 Acceptance Tests.--Sampling for all other Acceptance Tests shall be in accordance with Table V below.

Table V

Lot Size (number of Barometers)	Sample number selected	Acceptance number (defective)	Rejection number (defective)
9 to 12	4	0	1
13 to 17	5	0	1
18 to 25	6	0	1
26 to 35	8	0	1
36 to 50	10	0	1
51 to 70	13	0	1
71 to 100	17	0	1

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4.1.3.5 Acceptable Quality Levels.

4.1.3.5.1 Preproduction Samples.--All preproduction samples shall pass all tests to which subjected. Any failure shall be cause for rejection. Rejected samples shall be corrected and resubmitted until satisfactory models have been approved.

4.1.3.5.2 Visual and Mechanical Inspection.--Any barometer in the sample containing one or more visual or mechanical defects shall be rejected, and if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming items. The Government inspector shall again select and examine samples for each resubmitted lot to verify compliance with this specification.

4.1.3.5.3 Acceptance Tests.--Any sample failing to pass any of the applicable tests as designated in Table III shall cause rejection of the lot represented by that sample. Rejected lots may be resubmitted for inspection and tests, provided the manufacturer has removed or corrected all defective items.

4.1.4 Inspection Standards.--Concurrently with or prior to, the submission of preproduction samples as required in Paragraph 3.2 the contractor shall submit a complete description of the test standards he proposes to use in the inspection of barometers. This description shall be accompanied by three (3) certified copies of the most recent data taken on these standards in determining their accuracy. Full details concerning dates, methods used, and circumstances surrounding the taking of such test data shall be furnished. The contracting officer will advise the contractor of the approval or disapproval of the use of these instruments as standards for the inspection of the barometer on order.

4.1.4.1 Rechecking Inspection Standards.--The calibration of the inspection standards shall be rechecked throughout the test range for which they will be used at intervals not exceeding one (1) month during the time it is used as an inspection standard. The calibration of the inspection standard shall repeat within 0.2 millibars throughout the range at any time the inspection standard is rechecked.

4.1.5 Production Standard.--Upon satisfactory completion of the tests on preproduction samples, such changes shall be made as are necessitated by the tests. One finally approved sample shall be retained in the possession of the contracting officer until the end of production. The other finally approved sample shall be returned to the contractor's plant to be used by the inspector as a standard of workmanship and performance. The production standard shall be currently modified in accordance with any and all subsequently approved changes.

4.1.6 Facilities for Tests.--All materials and processes entering into the fabrication and testing shall be subject to the inspection of the authorized Government Inspector. The contractor shall furnish all necessary facilities and equipment for making the acceptance tests and inspections required by this specification, and shall carry out all tests under the supervision of the inspector. The test equipment shall be adequate in quantity to enable the inspector to keep up with production.

4.2 Test Procedure.

4.2.1 Pressure Variations.--The instrument shall not be subjected to pressure variation in excess of 15 millibars for a period of 15 hours preceding each test.

4.2.2 Rate of Pressure Change.--In making the tests at different pressures as required below, the rate of pressure change shall not exceed 10 millibars per minute.

4.2.3 Test Intervals.--In making tests at different pressures as required below, the test point intervals shall not exceed 10 millibars, and the corresponding test points on increasing pressures shall agree within ±1 millibar with related test points on decreasing pressures.

4.2.4 Errors.--When reference is made to an error permitted in a test, this error shall be defined as the difference between the indications of Barometer ML-448 and the corrected indication of the standard barometer when data is taken from both instruments simultaneously. When taking the reading of Barometer ML-448 the instrument may be tapped lightly. Pressures at test points shall be determined accurately to the nearest 0.05 millibar from the corrected readings of the test standard.

4.2.5 Visual and Mechanical Inspection.--The equipment shall be given a visual and mechanical inspection to determine the quality of material, workmanship, and compliance with the requirements of this specification.

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4.2.6 Scale Calibration at Room Temperature.--The instrument shall be maintained at ambient temperature 70°F ±5° at least 2 hours before testing. The pressure shall first be reduced from current atmospheric pressure to the lowest design pressure (910mbs), then increased to the highest design pressure (1060mbs) and finally reduced to current atmospheric pressure, in accordance with Paragraphs 4.2.2 and 4.2.3. The errors shall not exceed values specified in Table VI.

Table VI

(Values in millibars)

Test Point Values	Error
910 - 927	±1.00
927 - 961	±0.80
961 - 1030	±0.65
1030 - 1047	±0.80
1047 - 1060	±1.00

4.2.6.1 Change in Error between Test Points.--The errors as determined above shall not change in magnitude by more than 2 percent of the pressure change between any two consecutive test points.

4.2.7 Movement.--The movement of the indicating hand shall be noted in all tests specified in Paragraph 4.2. When the pressure is changed uniformly the indicating hand shall move uniformly. Irregular motion of the indicating hand shall be cause for rejection of the instrument.

4.2.8 Position Error Tests.--The indication shall be read while the instrument is in a vertical position. The instrument shall then be rotated successively into positions 45° forward and backward from vertical and 45° right and left of vertical and indications read in each position. Vertical readings shall be considered as those taken when the instrument is normally mounted against a vertical plane with the 1015 millibar graduation up. The difference in indication from vertical readings shall not exceed 0.3 millibars.

4.2.9 Temperature Cycle Test.--The barometer shall be subjected to a temperature cycle test. The instrument shall be maintained at a temperature of 160°F ±5° for a period of at least one (1) hour. It shall then be transferred in not more than five (5) minutes time to a chamber in which the temperature is -50°F. The instrument shall be maintained at this temperature for not less than one hour. This cycle shall be repeated at least four (4) times. At the completion of these cycles, the instrument shall be carefully examined. There shall be no evidence of damage to the instrument.

4.2.10 Pressure Cycle Test.--The barometer shall be subjected to a pressure of 500 millibars. It shall be held at this pressure for a period of one (1) hour and then the pressure increased to 1150 millibars for a period of one (1) hour. This test shall be repeated until a total of four (4) cycles have been made. After test of this paragraph the zero shift shall not exceed 0.5 millibars.

4.2.11 Friction.--The barometer shall be held in hand and read, without jarring or rotating, the instrument shall be slowly and carefully lowered three (3) feet. There shall be a perceptible movement of the indicating hand. The barometer shall then be raised three (3) feet. There shall be a perceptible movement of the indicating hand.

4.2.12 Hysteresis.--When tested in accordance with the procedure of Paragraph 4.2.6, the differences in errors for corresponding test points for ascending and descending pressure shall not exceed 0.4 millibars.

4.2.13 Recalibration.--The test specified in Paragraph 4.2.6 shall be repeated. The data obtained shall not differ from the original calibration test by more than 0.5 millibars.

4.2.14 Temperature Compensation.

4.2.14.1 Preproduction Samples.--Successive tests shall be conducted at a constant pressure within each of the following pressure ranges:

(a) 915mbs ± 5mbs
(b) 1000mbs ± 25mbs
(c) 1055mbs ± 5mbs

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At each pressure point, the temperature shall be changed from an ambient room temperature of 70°F ±5°F to each of the following temperatures:

(a) -20°F ±2°F
(b) +30°F ±2°F
(c) +120°F ±2°F

The barometer shall be held at each temperature for at least 1 1/2 hours before a reading is made, during which time the temperature shall not vary more than 5°F. Readings shall then be taken and the change in indications due to the temperature change shall not exceed 0.0090mbs per degree Fahrenheit.

4.2.14.2 Production Samples.--The above tests shall be conducted at any one constant ambient pressure only.

4.2.15 Salt Spray Corrosion Tests.--At least two sample aneroid barometers shall be subjected to the salt spray corrosion tests of Part G, Appendix II, of the General Specifications for Inspection of Material. Length of test shall be 48 hours.

4.2.15.1 Examination of Materials.--The external and internal structure of the barometers which have been tested in accordance with 4.2.15 shall be examined to determine evidence of corrosion. The presence of corrosion shall not cause the rejection of a barometer which otherwise complies with operation requirements of 4.2.18.

4.2.16 Vibration Tests.--The sample shall be subjected to the test specified in 4.2.16.1 through 4.2.16.4. The barometer shall be clamped in a normal mounting position and two other mutually perpendicular axes on a vibrating table that can be controlled within 10 percent of the specified frequencies and amplitudes. The vibration table shall be designed to provide approximately sinusoidal vibration at all specified frequencies and amplitudes.

4.2.16.1 Direction of Vibrations.--The barometers shall be vibrated successively over the specified ranges of applied frequencies in three mutually perpendicular directions; two directions of vibration shall lie in the plane of the barometer.

4.2.16.2 Rate of Change of Vibration Frequencies.--The frequency of applied vibration shall be varied uniformly at a rate of approximately 1 cycle per second per minute.

4.2.16.3 Amplitude of Applied Vibration.--The double amplitude (total excursion) of the vibration table shall be in accordance with table VII.

Table VII

Amplitude-frequency conditions

Frequency of vibration C.p.s.	Double Amplitude	
	Horizontal Inch	Vertical Inch
4 to 10	0.125	0.125
10 to 33	.060	.060

4.2.16.4 Ninety-minute Test.--The equipment shall be vibrated for at least 30 minutes in each of three directions specified in 4.2.16.1 (total of at least 90 minutes). The range of the applied frequencies shall be from 4 through 33 cycles per second.

4.2.17 Gun Blast Proof.--The barometer shall be constructed to withstand degree (1) gun blast proof of Specification MIL-E-2036.

4.2.18 Performance Recheck.--Upon completion of the tests of Paragraphs 4.2.15, 4.2.16 and 4.2.17, the performance tests of Paragraphs 4.2.8, 4.2.11, 4.2.12 and 4.2.13 shall be repeated.

5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing.--The equipment shall be packaged and packed in accordance with Specification MIL-P-6064 Group II. The contract shall specify whether packing shall be domestic or for overseas shipment. When overseas shipment is specified the equipment shall be packed in boxes in accordance with Specification JAN-P-106A.

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5.2 Marking.--All shipments shall be marked in accordance with MIL-STD-129.

6. NOTES.--Not applicable.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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